

Manual

Input/Visualization Functions for Process Data AIP-PDV 8.1

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1 Overview Input/Visualization Functions for Process Data

Purpose

The AIP features contained in this function package make it possible to display process data directly in production using shop floor terminals or data entry PCs.

Implementation considerations

You use the function package if you:

- Want to monitor the values entered via process data collection online at the terminal.
- Would like to visualize the system values in pointer instruments or trend lines.
- Would like to display system layouts using online measured values.

Integration

AIP offers the ability to connect to the data server used for input via the PDV visualization components and to display the values entered there online.

Features

Displays the input process values as a pointer or bar chart

- Illustrates the process trends using the latest input values
- Dialog controlled option to change target values manually
- Creates individual layouts independently or as a part of the HYDRA customizing process using the process visualization editor (MDS-PVE).
- Displays separate system layouts using the integrated measured values display and graphic display elements (additional license).

Additional licenses are needed in order to use the functions listed above.

2 Operation of AIP

2.1 Special control and display elements within AIP

Tables

Tables are displayed in a uniform way within AIP. This affects the basic display (workplaces, operations, ...) as well as the selection lists of posting dialogs.

Filter	
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE

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Provided that information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. By clicking/touching the user can directly switch to another page.

An operation may be selected using the mouse, touch screen, keyboard (arrow keys: '↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Please find the following example: When an operation is logged on, those operations may be selected that are included in the sequencing list or that are planned for the corresponding workplace or group. However, when operations are interrupted, only running operations may be selected.



Scrolling page by page (up or down) in the table.



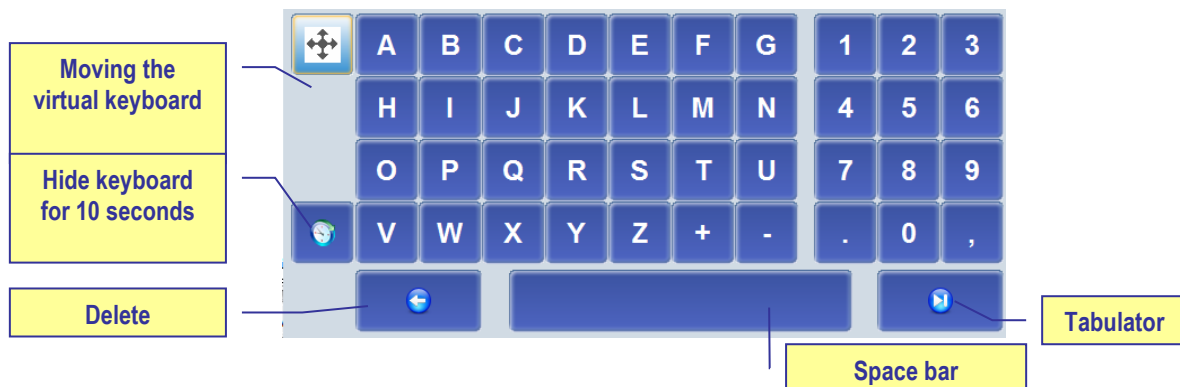
Scrolling to the left or right. Only those buttons are activated that are reasonable for the current situation. This figure shows that scrolling to the left has been deactivated.

Filter A

“table filter” may optionally be displayed (customizing). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is case-insensitive.

Virtual keyboard

The virtual keyboard allows for data to be entered manually via touch screen or a connected mouse. To make it easier for inexperienced users to find the required keys, the numeric key pad is organized like the telephone and letters are aligned in alphabetical order. Consequently, both differ from the computer keyboard which usually is aligned in the “QWERTZ keyboard layout”. The virtual keyboard is displayed automatically as soon as an input field is focused.



The driver needs to be configured respectively for the touch screen to be able to move the keyboard (settings in the control panel of the terminal/PC)! The virtual keyboard only supports the characters "0" - "9", "A" - "Z" and "+", "-", ".", and ",". Other characters or languages are not supported. It is recommendable to use an additional keyboard if texts in other languages have to be entered.

The start position of the virtual keyboard can be defined by a setting in the configuration file keyboard.ini. Subject to the screen resolution, the parameters xpos= and ypos= need to be enabled in the configuration file.

If the virtual keyboard is not to be shown in general, the parameter -t needs to be included in the parameter bar parameters= of the configuration file ctaip.ini.

Date display

AIP supports a country-specific date format in dynamic dialogs. This can be configured in the "control panel", "regional settings", "short date" dialog of the terminal/PC. The following has to be taken into account in this context:

- Years are always four characters long.
- Months and days are always 2 characters long.
- “-”, “/” and “.” are allowed separators
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- | | |
|-----------------------|------------|
| • English(USA) | MM/DD/YYYY |
| • Danish | MM-DD-YYYY |
| • Customer-specific 1 | YYYY-MM-DD |
| • Customer-specific 2 | YYYY/MM/DD |
| • Customer-specific 3 | MM/YYYY/DD |

Please note

If the date format does not correspond to conventions a note appears when the program is started and the date format is set to MM/DD/YYYY.

The year is displayed only by two characters in the status bar.

2.2 General description of the posting process with AIP

In general, posting dialogs are divided into several visual views at AIP. These views (partial dialogs) cover the entire screen and only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. This process is described by way of the following example (interrupt operation). The other dialogs can be operated in the same way.

The “interrupt operation” function is executed. This task is started by clicking the “interrupt operation” function from the second toolbar:



The “interrupt operation” dialog opens and the first view is displayed. The function that is currently being executed (in this case: interrupt operation) is shown in the header.

The first view “enter quantities” provides the user with the possibility to enter the produced yield or scrap quantities. The virtual keyboard is shown or hidden automatically, subject to the active input field.

Quantities can be entered using the virtual keyboard or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). Once all values have been entered in the first view, the next view can be opened by clicking the “next” button.

The “cancel” button is displayed in all partial dialogs and allows for the entire posting dialog to be cancelled/closed at any time.

The next view can be opened either by clicking the “next” button or by clicking another tab (in our example: “select status” or “confirm”). Please note in this context, that no view can be skipped when the views are navigated bottom up (view 1 → view 2 → view 3). This means: if you are in the first view (enter quantities) and you click the third view (confirm), the second view (select status) will be displayed first.

Vice versa, when navigating top down (e.g. from the “confirm” view to the “enter quantities” view), every view may directly be opened by clicking at it. In this case, views are actually skipped. But the “back” button also allows for the views to be opened one after the other (top down).

As long as the dialog has not been confirmed, entered data may be changed at any time by scrolling back and forth.

Interrupt operation 4199LL120010 Assembly at workplace 406457A

1 Enter quantities
2 Choose status
3 Confirmation

Workplace 406457A MOUNT1

Actual status PRODUCTION since .. 18:40:41

Filter field for the list

Filter

Status	Status
1	PRODUCTION
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE
501	COLOUR SAMPLE

Status list

New status 6 START UP Estimated duration 0 Minutes

Comment

Cancel
Back
Go on

07/12/12 23:14:58

The workplace status that is to be set, once the operation has been interrupted, is determined in the second view “select status”. This status may be chosen from the displayed status list. This list can be restricted using the “filter” field. Once the required values have been entered, the next view can be opened by clicking “next” (in our example it is the last view).

Interrupt operation 4199LL120010 Assembly at workplace 406457A

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0010 Assembly

Prod.Order	4199LL12	Target qty.	30000 PCS
Article	34364-647-345	Yield	6833 PCS
	Seating Plate	Scrap	333
		Completion	22%

Workplace 406457A MOUNT1

New status START UP

Workplace data

Entered quantities

Yield 22
Scrap 7

Quantities posted for the OP.

Staff Number

Input field for the badge number

Cancel

Back

Interrupt OP

07/12/12 23:16:40

The partial dialog “confirm” shows a summary of all values entered so far in the dialog. Provided that the user agrees with the entered data, the “interrupt operation” dialog can be confirmed, once the badge number has been entered. Then the dialog including the data is sent to the server and posted.

If an input field of the dialog is not filled out properly (e.g. a mandatory field is empty) the field is highlighted in red in the corresponding view and focused to enable the user to directly correct the field content.



If a workflow dialog is opened it may directly be exited by clicking the ESC button. This is also the case, if the virtual keyboard is opened. Thus, the ESC button cannot be used to close the virtual keyboard.

3 PDV Functions - Process Visualization in AIP

Purpose

As a part of the basic HYDRA-PDV functionality, the AIP terminal program transfers inspection plans to the respective interface via MWP2 or a driver DLL. Vice versa, the AIP terminal then transfers incoming measured values to the HYDRA server.

The process visualization function allows to monitor the status of measurement channels at the AIP terminal.

3.1 Display of measuring channels

By pressing the "PDV" button in the "workplaces" section of the AIP terminal it may be switched to the respective display. The measuring channels of the machine selected in the basic screen are shown. The workplace may also be changed in the HYDRA-PDV view. An empty HYDRA-PDV dialog will be opened if no measuring channels are configured for the selected machine.

The value range of all displays corresponds to the tolerance range of the respective measuring channel configured within the inspection plan.

Different modes are available to display measuring channels. The mode may be chosen from the lower button bar.



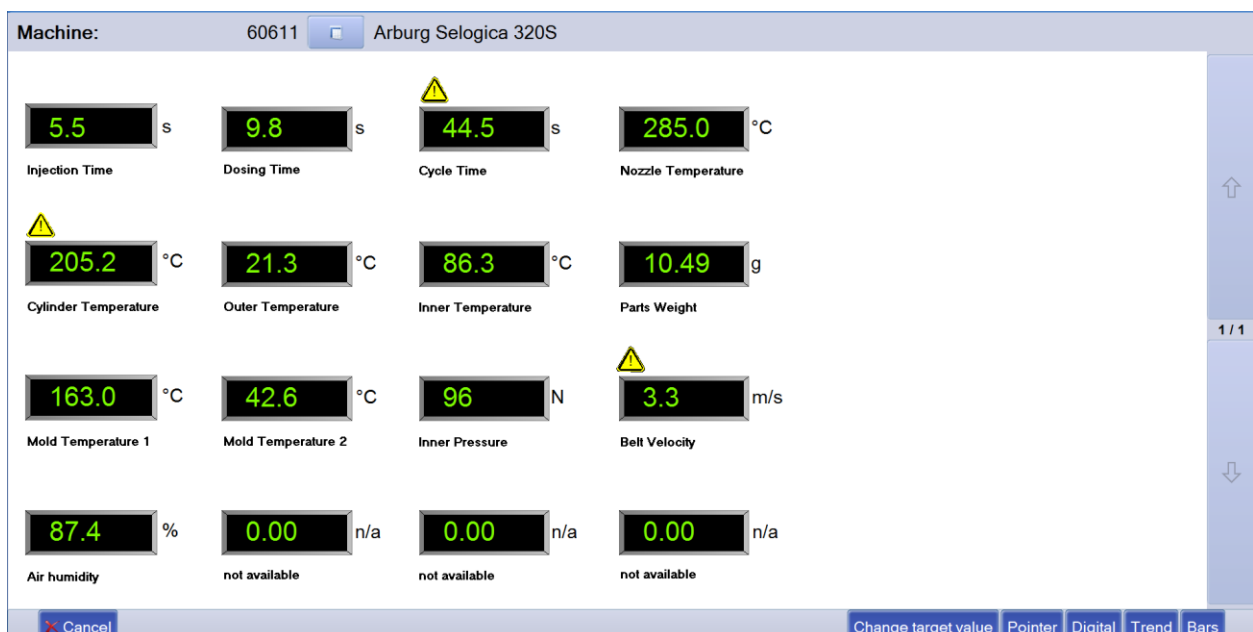
Pointer display

Measuring channels are displayed as pointers. All of the up to 16 configurable measuring channels of a machine are also displayed at the same time.



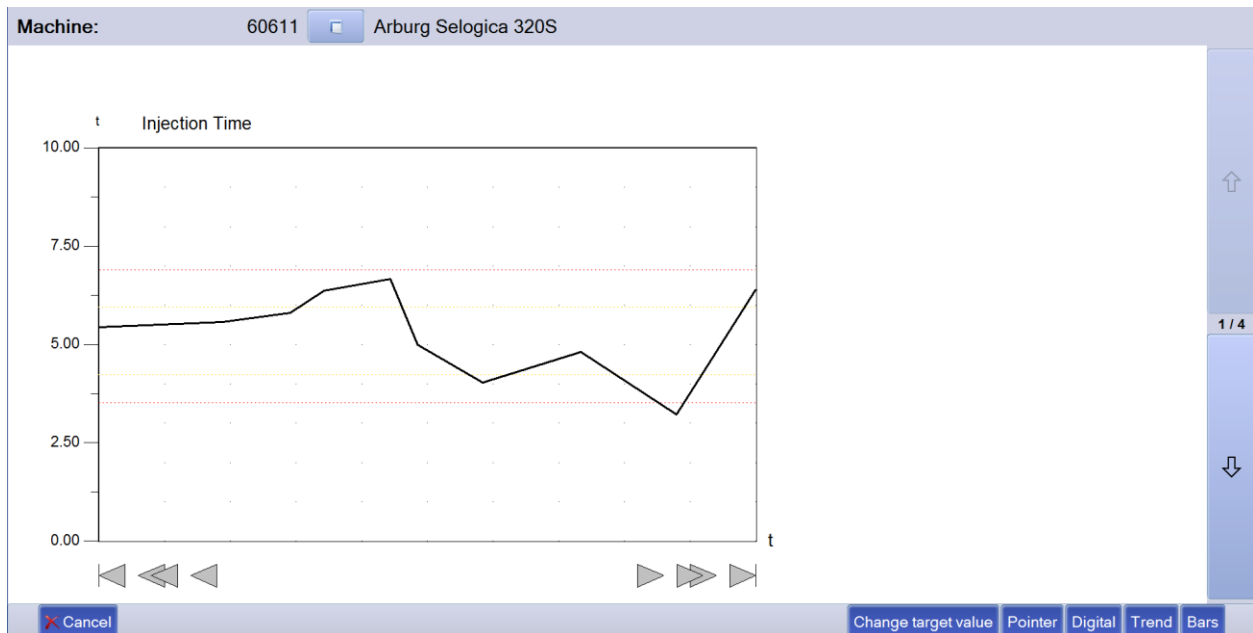
Digital representation

Measuring channels are represented as digital values. All of the up to 16 configurable measuring channels of a machine are also displayed at the same time.



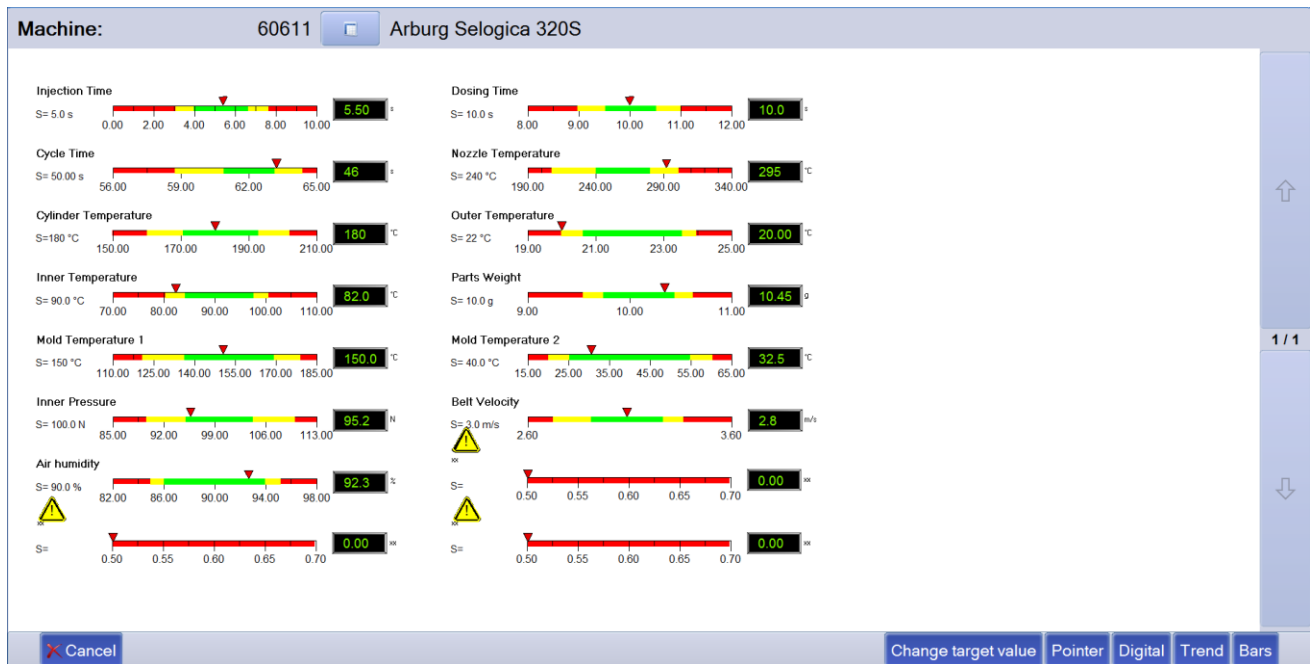
Trend display

The progress including a corresponding legend is shown for each process parameter in a graphic. The other process parameters can be reached by the arrow keys at the right margin of the screen.



Bar display

Measuring channels are displayed as bars. All of the up to 16 configurable measuring channels of a machine are displayed at the same time:



3.2 Change default value

Default values represent, among other things, a decisive factor when process data are displayed. They may be changed by clicking the “change default value” button.

Posting procedure

The required workplace has to be selected, before changing default values.

Starting of the “change default value” function

The “change default value” button is to be clicked. As soon as the function has been started, the user is navigated through the dialog. The workplace has already been defined.

Select process parameter

Change target value for process parameter at machine 60611

1 Choose process parameter 2 Change target values

Machine 60611 ARB-320S

Actual status UNDEFINED since 11.10.2010 14:00:00
Duration 15022:36:0

Filter

Cancel Go on

The required process parameter, the default values of which have to be changed is to be chosen from the available list.

Change default values

Change target value for process parameter at machine 60611

1 Choose process parameter 2 Change target values

Operation 0100 InjectionMolding

Prod.Order S6100328 Article SAL-OSI011-K-9001
Mirror housing left, crem

Process parameter ()

Upper tolerance limit
Upper process interv. limit
Target value
Lower process interv. limit
Lower tolerance limit

New target value

Actual target value

Staff Number

Cancel Back Change target values

The individual values that are to be changed are entered in the dialog that opens. This dialog shows the previous values as well as the new default values. The following values are concerned:

- **Upper tolerance limit**

- *Upper process action limit*
- *Target value*
- *Lower process action limit*
- *Lower tolerance limit*

Badge number

The badge number of the person changing the data is to be entered here.

Confirmation of “change default values”

The default values are updated in the system by confirming the dialog. They in turn affect HYDRA-PDV display at the AIP terminal.

4 Local Configuration File ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the c:\ctaip directory.



Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

4.1 Basic configuration

Entry	Comment
Section [system]	
Usr=21	Distinct terminal number
Hypath=d:\hydra\ Hypath=/usr/hydra/	HYDRA server path: Windows NT: -> DOS notation, the drive is the local drive of the server on which HYDRA or xMES is installed Unix: -> Unix notation
Hostname=192.9.200.24	Internet address of the server
Offlinetimeout=600	In offline mode, the interval after which online access should be attempted the next time. The interval is specified in seconds
Showcursor=on	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
Loadfile=ctnet\win\ctaip.txt	Configuration file for downloading the application from the server. The path is relative to the server directory (i.e. within "hypath")
Watchdog=on	ON: Watchdog is activated OFF: Watchdog is not activated
Demo=off	'on': Offline demo mode; always off in the production environment
parameters=-t	The -t parameter switches off the virtual keyboard
TMOUT_C=xxx	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_S=xxx	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds

Entry	Comment
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	Configuration of customized barcode prefixes.
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
Com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to the connected devices MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
ZAEHLER= 1 2 3 4 5 6 7 8	Assignment of physical inputs of the MSS (machine interface) to logical counters (ZAEHLER) according to configuration: The first connector (labeled “0” on the MSS) corresponds to the logical counter no. 1 Please note: MSS1 has only 8 inputs If digital inputs are also to be used with MSS1, configuration should be changed as follows: ZAEHLER= 1 2 3 4 IN= 5 6 7 8
IN= 9 10 11 12 13 14 15	Assignment of physical MSS inputs to logical inputs as per configuration: The ninth connector (labeled “8” on the MSS) corresponds to the logical input no. 1
CHARGE= 5 6	For batch recording: Inputs for automatic batch changes In this case, the connectors 5 and 6 are the inputs for the automatic batch change.
MSSZyklusBerechnung=ON MSSZyklusReferenz=0	Activates reading out of cycle time values from the machine interface (MSS). Reference for cycle calculation (smallest time unit of the MSS). The default value is 0, if the parameter is not specified. 0 corresponds to 100 ms. 2 corresponds to 20 ms.
CalculateCycle=ON	The terminal itself calculates the actual cycle. If the connected control does not provide a determined actual cycle, then a calculated actual cycle can be displayed: This calculated value is also available for DS100 terminals. Please note: The calculation cannot provide exact values.
WochenEnde_ProdCheck=ON	This function prevents the weekend automatic from affecting the “production” status and the workplace from being set to status 999. ON is set by default In case WochenEnde_ProdCheck=OFF, the automatism switches to status 999.

Entry	Comment
sFrom999ToNotAttributed=OFF	Only affects HYDRA-MDE machines with operation mode = "no monitoring". If the weekend automatic function is enabled this option prevents the machine from switching to the "not assigned" status when the weekend automatic ends. Reasons: The "not assigned" status may not be set manually for machines that are configured with the "no monitoring" option. The "not assigned" status is normally only set for machines with operation mode = "cyclic monitoring" or "monitoring by operating signal".
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be called to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [PDV]	
Modus=PDV Modus=PDV,BDE	Operation as interface IOP $\leftrightarrow$ DOS terminal (standard) Operation as shop floor terminal with PDV
Terminals=121,122,123	CTDOS terminals "connected" by LAN. Only required with Mode=PDV
PDVTerminalDir=c:\hsrv\spool\ PDVIOPTDir=c:\IOPSim\	Directory for communication with CTDOS terminals Directory for communication with the IOP
For debug purposes only: InfoFenster=100 SlowDown=600	Number of lines of the "current" window (presentation of last PDV actions) Slow down, to make events "visible"
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.

Entry	Comment
BarcodeNest=	This field specifies which acronym is entered into the cavity number field by the scanner.
BarcodeNumm=	This field specifies which acronym is entered into the number field by the scanner.

5 Central Configuration File hytnrcfg.ini

This file includes different configurations for all or single terminals at a central place.

Each section is available in a generally accepted version
[section 0].

However, entries included in this section can be overwritten by entries in a terminal-specific section
[section <TNR-USER>]

<TNR-USER> = HydraUser = Terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/HYDRA User



The hytnrcfg.ini file is loaded from the server every time the terminal is started.

Section / Entry	Comment
	→→→
[Tnr configuration 0]	
FollowExternStatus=on	Transfer of machine statuses when reloading machine list Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
InstallFonts=on	If this is set to "off" fonts will not be installed during the restart. ON=DEFAULT
OnlyInstallFontsAfterDownload=false	"InstallFonts=on": If true then fonts will only be installed directly after a download. If false then fonts will be installed every time the terminal is restarted. (false = DEFAULT)
InstallTvicport=on	If "off" the LPT driver "tvicport.sys" will not be installed. It is required for HYDRA-ZKS. ON = DEFAULT
[Terminal->USR 0]	

AttachedApplication=First	This configuration checks whether or not an application is connected in Windows that matches the file extension of the document to be displayed from the OP info dialog. If there is such an application, it will be used for displaying the document. If there is no connection, viewers configured in ctaip.ini (→ [ext. software]) and internal viewers will be used. In case, an extension is completely unknown it is attempted to display it as text Different settings may be configured: First → search for connected application first AfterUserViewer → If a UserViewer is configured this one overrides the connected application (also applies for ExcelViewer, WordViewer and PowerpointViewer) Last → Only if no ctaip.ini assignment is found for the file extension, then the connected assignment will be searched for (default). Off → Connected application is never searched.
HTTPBrowser=standard	Viewing of documents (via OP info): If documents are configured with a path of the <u>type</u> "http", the file will not be downloaded to the terminal, but the link will only be transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows will be used.
SupressErrorMessage=70012	Suppress message "material is not planned"
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field "user" can be edited in the terminal (by default: no editing) true → activates keyboard input for the "user" field in the terminal

Transparency=255	The signature dialog can also be transparent. 255 → Signature dialog is 0 % transparent (not transparent) 1 → Signature dialog is 99% transparent (maximum transparency) (Default = 155) Available as of CTAIP (V# 2.0.2.25) / CTWIN (V# 7.2.5.99)
ShowPosition=TR	The position of the signature dialog can be adjusted as follows: TL Top – Left TM Top – Middle TR Top – Right ML Middle – Left MM Middle – Middle (Default) MR Middle – Right BL Bottom – Left BM Bottom – Middle BR Bottom – Right Available as of CTAIP (V# 2.0.2.25)
USE_SERVICE_ACCOUNT=1	0 (default) SSO: do not use ServiceAccount (requires the terminal to be started with the "user" domain (SSO)). Please note: ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY	REPORTING_USER_READONLY The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field is read-only. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. REPORTING_USER_CHANGEABLE The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field can be modified. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible.

SIGNATURE_1_LOGON_TYPE=HYDRA	"" / Not set / "EMPTY" There is also an alternative login procedure. HYDRA The tab identifying users via the Windows user is blocked. The HYDRA user must be used for identification purposes. This requires, however, that in the HYDRA HR master all users logging in are created and that the "SSO" option is not set. Otherwise, successful authentication is impossible. ACTIVEDIRECTORY The tab identifying users via the HYDRA user is blocked. The Windows user must be used for identification purposes. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. MIXED_BUT_UNIQUE Either the HYDRA or Windows login procedure is available, subject to whether or not the "SSO" option is set for the registered user in the HYDRA HR master. "SSO" enabled → Windows only "SSO" disabled → HYDRA only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures with the terminal in the area of quality data collection.

5.1 Layout configuration

Entry		Comment
Section and/or	[terminal configuration 0] [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE		In case of an error in reading the clock (e.g. after coming out of standby mode), this configuration makes sure that the time is accepted without having to confirm a dialog. Afterwards the terminal time will be synchronized with the server time using a PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON		As of ctaip V# 2.0.2.23 Prevents the warning after restarting the terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default = OFF)

Entry	Comment
NetRuntimeMode=2	As of ctaip V# 2.0.2.50: Alternative calculation of the target quantity since logon: The net run time is not calculated from the times when the production lock is enabled (PSperre=green) but only from the shift times less the shift breaks. Consequently, it can also be displayed, even if the terminal program has been restarted.
Section [QRD-PRINTER->TICKET 0] ;(general configuration) [QRD-PRINTER->TICKET 2xxx] ;(2XXX configuration for a specific terminal)	
COMPLETE-ABSENCE-OF-LOCAL-MNR- DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured <Events>, if it is not available locally => This configuration might be required/necessary for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR- DATA-FOR-EVENT=< Events >	Reloads the order row for the configured <Events>, if it is not available locally ➔ This option has been implemented to access order data within the master data, e.g. when logging an order on.
COMPLETE-...-EVENT=< Events > COMPLETE-...-EVENT=#ALL# COMPLETE-...-EVENT=A_AN A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN A_P_AN> restricts reloading of information to the specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.